



For queries:

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Round 1 Instructions (Due date: 03.08.2025)

- 1) Each team is expected to annotate the following dataset using CVAT (offline).
 - i) Crop disease dataset (50 images):
<https://drive.google.com/drive/folders/1-mgBbJnEOSeWrKsgmn4aDWbHKaUAeGrD?usp=sharing>
 - ii) Crop insect dataset (50 images):
https://drive.google.com/drive/folders/1szQoqBQdBH4xm8Yo_FxQ34xVGpJOnTgl?usp=sharing

2) Each team is expected to implement a multimodal:

→ **Refer Table 1:Round-1 Implementation details**

3) Each team is expected to submit the **round-1 document** to ***shunmugaperumal.p@vit.ac.in*** with the following contents on or before **03-AUG-2025 (Sunday)**:

- i) CVAT (offline) setup procedure.
- ii) Annotation techniques used.
- iii) Annotated dataset (*Attach Google Drive link in the round-1 document*).
- iv) Architecture of the multimodal with
 - short & clear description.
 - Pictorial representation (Clear Hand-drawn pictures also fine)
- v) Implementation codes of the multimodal in an organized manner.
- vi) Procedure to run the entire application (end to end) in the terminal.
 - GUI is not required.
- vii) Output screenshots
 - Project/tabulate all possible highlights (model performance parameters) of your multimodal.

Note:

→ **Model accuracy is NOT the round-1 selection criteria.**

→ The main objective of this round-1 process is to select suitable teams for the Round-2 of the Hackathon by assessing the following skills:

- Linking/integrating different models (YOLOv8s-seg, YOLOv8s, TabNET) together to form a multimodal.
- Documentation skills.

Table 1: Round-1 Implementation details

1	Annotation Tool to be used	CVAT (offline)	Teams are expected to explore the full potential of the CVAT (offline)
2	Annotation/segmentation	- Bounding box (crop Insect images) - Segmentation (crop disease images) - Segment disease regions on leaves	
3	Augmentation (after annotation):	- Normalization - Rotation - Flipping - Color Jittering	Participants are expected to use other possible augmentation techniques.

		Contrast enhancement	
4	Multimodal implementation	Input 1: Image disease Input: - Crop disease Image will be the input. - Crop disease must be detected (output-1).	YOLOv8s-seg model trained on segmented dataset must be used to detect crop disease.
		Input 2: Image insect Input: - Crop insect Image will be the input. - Crop insect must be detected (output-2).	YOLOv8s model trained on segmented dataset must be used to detect crop disease.
		Input 3: Text input from CSV file: - A list of questions (symptoms of crop disease) along with the corresponding crop disease images (from the sample collection) will be displayed to the farmers. - Farmers will look into the crop disease images displayed by the system and answer the questions (Yes/No type inputs). - Eg: Question by the system: Do you see a brown color infection on	- TabNet model. - For TabNet, you can generate synthetic data with AI for the given questions and train.

		the leaf? ○ Answer by the Farmer: Yes ● Farmer's input (text type) will be taken as input by the TabNet model to predict the presence of the crop disease (output-3). ● Refer the csv file for Farmer's text input (crop_disease_characteristics.csv) to TabNET.	
		Input 4: Text input from CSV file: ● A list of questions (symptoms of crop insect) along with the corresponding crop insect images (from the sample collection) will be displayed to the farmers. ● Farmers will look into the crop insect images displayed by the system and answer the questions (Yes/No type inputs). ○ Eg: Question by the system: Do you see a	● TabNet model. ● For TabNet, you can generate synthetic data with AI for the given questions and train.

		green color larva on the crop leaf? ○ Answer by the Farmer: Yes ● Farmer's input (text type) will be taken as input by the TabNet model to predict the presence of the crop insect (output-4). ● Refer the csv file for Farmer's text input (crop_insect_characterists.csv) to TabNET.	
		● Fusing the outputs of all models together to get the final conclusion:	● The multimodal pipeline must fuse all outputs (output-1, output-2, output-3, output-4) to get the final conclusion (final_output). ● Final_output: ■ Crop disease present or not present ■ Crop insect present or not present

Summary Table:

Model	Task	Type	Suitability
YOLOv8s-seg	Crop disease (segmentation)	Image segmentation	✓ Excellent
YOLOv8s	Crop insect (bounding box)	Object detection	✓ Excellent
TabNet	Farmer Q&A on symptoms (text)	Tabular classifier	✓ Excellent